

Written Homework 7

● Graded

Student



Total Points

9 / 12 pts

Question 1

Question 1

2 / 3 pts

– 0 pts Correct

– 1 pt computational error

✓ – 1 pt wrong answer

– 2 pts No work

💬 5^x should have another power of 2^x

Question 2

Question 2

2 / 3 pts

– 0 pts Correct

✓ – 1 pt Part a answer wrong

– 1 pt Part b answer wrong

💬 $\ln x \neq 1/e$

Question 3

Question 3

3 / 3 pts

✓ – 0 pts Correct

– 0 pts Click here to replace this description.

Question 4

Question 4

2 / 3 pts

– 0 pts Correct

✓ – 1 pt Wrong sign on t squared

– 1 pt No t-0

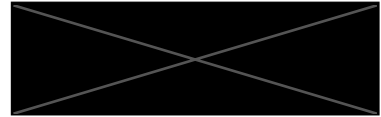
– 1 pt Wrong coefficient on t squared

– 1 pt No t squared term

– 2 pts No answer

No questions assigned to the following page.

Written Homework 07: §3.6, §4.2



- **Instructions:** Work out solutions on scratch paper before writing the solutions you submit. Your solutions must completely and clearly justify your final answers for your work to earn credit. Write legibly. You may not receive credit for work that is illegible or unintelligible. Write full sentences whenever possible. Leave plenty of space on the page so that graders can easily write comments. **Submit your answers by Friday October 25, 2024.** Late submission, free of penalty, will be accepted for 48 hours. Portions of assignments uploaded after the hard deadline will not be accepted for credit. Your work must be submitted through Gradescope for it to be graded.
- **Honor Code:** You should work individually at first. After you have given the questions a good amount of independent thought, then you are encouraged to collaborate and work together with your Math 101 peers. You may also ask for hints or suggestions (but not solutions) from other academic resources at Rice (e.g. instructors, peer tutors, other students). However, you are not allowed to look up solutions in any written form; in particular, you are not allowed to look up solutions to problems in a solution manual or other online sources, or to seek help from AI systems.

1. Find the derivative of $\left(\frac{5^x}{1 + \ln(x)}\right)^{2x}$.
2. Consider the function $f(x) = \ln(x)$.
 - (a) Find the Linearization of f at $x = 10$.
 - (b) Use $\ln(10) \approx 2.303$ to approximate $\ln(11)$.
3. Suppose that f is a differentiable function. If $f(2) = 4$ and the derivative satisfies $3 \leq f'(x) \leq 5$ for all x , then how large and how small the value of $f(8)$ could be?
4. Suppose that an object is falling freely from rest with downward acceleration given by 9.8 m/s^2 . Find the formula for the position function $s(t)$ if the initial height was 14 meters.
5. Consider the function $f(x) = \sin(x)$.
 - (a) Apply the Mean Value Theorem on the interval $[0, x]$ for $x > 0$.
 - (b) Use $|\cos(x)| \leq 1$ to show that for $x > 0$ one has $|f(x)| \leq x$.
 - (c) What if $x < 0$?

Questions assigned to the following page: [1](#) and [2](#)

1. Find the derivative of $\left(\frac{5^x}{1+\ln(x)}\right)^{2x}$.

$$\begin{aligned}
 \left[\left(\frac{5^x}{1+\ln(x)}\right)^{2x}\right]' &= \left[e^{\ln\left(\frac{5^x}{1+\ln(x)}\right)^{2x}}\right]' \\
 &= \left[e^{2x \ln\left(\frac{5^x}{1+\ln(x)}\right)}\right]' \\
 &= \left[e^{2x \ln\left(\frac{5^x}{1+\ln(x)}\right)}\right] \left[2x \ln\left(\frac{5^x}{1+\ln(x)}\right)\right]' \\
 &= \left[e^{2x \ln\left(\frac{5^x}{1+\ln(x)}\right)}\right] \left[2 \ln\left(\frac{5^x}{1+\ln(x)}\right) + 2x \left[\ln(5^x) - \ln(1+\ln(x))\right]\right]' \\
 &= \left[e^{2x \ln\left(\frac{5^x}{1+\ln(x)}\right)}\right] \left[2 \ln\left(\frac{5^x}{1+\ln(x)}\right) + 2x \left(\frac{1}{5^x} \cdot 5^x \ln(5) - \frac{1}{1+\ln(x)} \cdot \frac{1}{x}\right)\right]' \\
 &= \left(\frac{5^x}{1+\ln(x)}\right)^{2x} \left[2 \ln\left(\frac{5^x}{1+\ln(x)}\right) + 2 \left(\frac{\ln(5) \times (1+\ln(x)) - 1}{1+\ln(x)}\right)\right]' \\
 &= \boxed{2 \left(\frac{5^x}{1+\ln(x)}\right)^{2x} \left(\ln\left(\frac{5^x}{1+\ln(x)}\right) + \frac{\ln(5) \times (1+\ln(x)) - 1}{1+\ln(x)}\right)}
 \end{aligned}$$

2. Consider the function $f(x) = \ln(x)$.

(a) Find the Linearization of f at $x = 10$.

(b) Use $\ln(10) \approx 2.303$ to approximate $\ln(11)$.

$$(a) \left. \frac{dy}{dx} \right|_{x=10} = \frac{1}{10}$$

$$\text{Tangent line @ } x=10: y = \frac{1}{10}x + \ln(10) - 1$$

$$\begin{aligned}
 L(10) &= f'(a)(10-a) + f(a) \\
 &= f'(e^2)(10-e^2) + f(e^2) \\
 &= \frac{10-e^2}{e^2} + 2 \\
 &= \frac{10}{e^2} + 1 \\
 &\approx \boxed{2.35335}
 \end{aligned}$$

$$\begin{aligned}
 (b) L(11) &= f'(10)(11-10) + f(10) \\
 &= \frac{1}{10} + \ln 10 \\
 &\approx 0.1 + 2.303 \\
 &\approx \boxed{2.403}
 \end{aligned}$$

Questions assigned to the following page: [3](#) and [4](#)

3. Suppose that f is a differentiable function. If $f(2) = 4$ and the derivative satisfies $3 \leq f'(x) \leq 5$ for all x , then how large and how small the value of $f(8)$ could be?

Since f is a differentiable function, it is also a continuous function.

$f(x)$ is continuous on $[2, 8]$ and differentiable on $(2, 8)$, so by the Mean Value Theorem, there is a c such that $f'(c) = \frac{f(8) - f(2)}{8 - 2}$.

$$\text{Minimum Value of } f(8): 3 = \frac{f(8) - 4}{6} \rightarrow f(8) = \boxed{22}$$

$$\text{Maximum Value of } f(8): 5 = \frac{f(8) - 4}{6} \rightarrow f(8) = \boxed{34}$$

4. Suppose that an object is falling freely from rest with downward acceleration given by 9.8 m/s^2 . Find the formula for the position function $s(t)$ if the initial height was 14 meters.

$$\begin{aligned} a(t) &= 9.8 \text{ m/s}^2 \\ v(t) &= 9.8t \text{ m/s} \\ s(t) &= 4.9t^2 + 14 \text{ m} \end{aligned} \quad \begin{array}{l} \swarrow \text{Antiderivatives} \\ \downarrow \end{array}$$

5. Consider the function $f(x) = \sin(x)$.

- Apply the Mean Value Theorem on the interval $[0, x]$ for $x > 0$.
- Use $|\cos(x)| \leq 1$ to show that for $x > 0$ one has $|f(x)| \leq x$.
- What if $x < 0$?

(a) $f(x)$ is continuous on $[0, x]$ and differentiable on $(0, x)$, so by the Mean Value Theorem, there is a c such that $f'(c) = \frac{f(x) - f(0)}{x - 0}$.

$$f'(c) = \frac{f(x) - f(0)}{x - 0} \rightarrow \cos(c) = \frac{\sin(x)}{x}$$

(b) $|\cos(x)| \leq 1$

By our application of MVT in part (a), there exists $\cos(c) = \frac{\sin(x)}{x}$.

$$\text{For } x > 0 \rightarrow |\cos(x)| \leq 1 \rightarrow \left| \frac{\sin(x)}{x} \right| \leq 1 \rightarrow |\sin(x)| \leq x \rightarrow |f(x)| \leq x.$$

$$(c) \text{ For } x < 0 \rightarrow |\cos(x)| \leq 1 \rightarrow \left| \frac{\sin(x)}{x} \right| \leq 1 \rightarrow |\sin(x)| > x \rightarrow |f(x)| > x.$$

For $x > 0$, $|f(x)|$ will always be positive and x will always be negative, so $|f(x)|$ is always greater than x .